

Exploration for a Framework to Preserve Implicit Knowledge for Game Development Organizations

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This project proposes a work-in-progress for establishing a framework for visualizing implicit knowledge that is generated in an organization focused primarily on the development of video games. This project is exploratory in nature and thus takes a dive into understanding how current techniques, such as video game postmortems, contribute towards conversion of implicit knowledge into explicit knowledge, and that in turn increases the organizational knowledge of the organization. Game development, in general, can be a very unpredictable process, with many variables coming into play during the development cycle. As a result, there may be a lot of changes, additions, and subtractions to the initial idea that was proposed. It is of utmost importance that these changes be documented in a way that can enable current and future members of the organization to draw implications and conclusions from these development cycles. Many of these postmortems are highly informal in nature, and usually these are a result of various interviews and interactions with the leaders and experts of the teams involved. Thus, this informal structure might lead to a varied and disparate body of literature, which may not always be the most efficient way to acquire the knowledge of the past. In this paper, we try to establish a way of extracting relevant information from this source and try to implement a visual mode for understanding what insight we can gain from it. The finding suggests that it is possible to develop a visual interpretation of postmortems and similar implicit knowledge-based texts, however there are several steps that need to be satisfied first. Development of a proper knowledge base, focused for the domain of game development, is crucial to be able to further work on it. This means development of a proper taxonomy and ontology, that addresses the concepts in the field of game development. Through properly framed knowledge bases, knowledge graphs can be created which have proven to be a primary factor in the success of an organization. Next, the introduction of an expert system, using the base of explicit knowledge (internal documentation), would enable proper utilization, visualization, and recall of this knowledge base. Finally, an emphasis on a Community of Practice approach for the entire framework would allow developers to share and transfer knowledge among themselves, while using the existing documentation as the base, in the process turning static documentation into a more dynamic and interactive source to draw knowledge from.

CCS CONCEPTS: • Information systems • Information systems applications • Decision support systems • Expert systems • Information systems • Information systems applications • Digital libraries and archives • Human-centered computing • Visualization • Visualization application domains • Information visualization

Keywords: Implicit knowledge, game development, knowledge base, knowledge graph, expert systems, community of practice, learning framework, organizational knowledge

Introduction

Knowledge is one of the primary aspects of the life of humans. Throughout the entirety of a human's life the amount of knowledge that gets acquired varies from stage to stage, as well as state to state. What remains constant however, is the process of the continual acquiring, processing, disseminating of knowledge. This is the fundamental aspect of learning and progressing in life. Knowledge can encompass any information that the human can acquire and comprehend. Merriam-Webster dictionary defines knowledge as “the fact or condition of knowing something with familiarity gained through experience or association” [1] Experience is a major part of knowledge – this concept applies towards organizations as well. This leads to the creation of the concept of “organizational knowledge”. According to Tsoukas and Vladimirou (2002) [2] organizational knowledge can be defined to be the knowledge generated within an organization due to the members carrying out their work and experiences, contributing towards a collective understanding as well as influencing decisions for the current and future workings of the organization.

Nonaka and Takeuchi [3] in their research stated the different spectrums of knowledge that could exist due to the environment in which the company evolved. According to them there exists a need to understand knowledge creation in organizations. This research led to the categorization of knowledge within organizations into two categories: tacit and explicit. Tacit or implicit knowledge was stated to be experiential i.e., the more experience was gained, the more scope for generation of tacit or implicit knowledge. Tacit or implicit knowledge was also stated to be rooted in practice. Compared to explicit knowledge (which was more theoretical and based on rational processes), tacit or implicit knowledge was formed through mental models, schemas, and experiences. Their argument laid out the fact that for Japanese companies to make a mark in the international market, tacit (or implicit) knowledge was an important factor. For the rest of the paper, tacit and implicit knowledge will be used interchangeably. [20]

Game development is a field that deals with many unique projects and tasks and is a rapidly growing industry. According to Woo et al. (2004) [4] due to the highly unique nature of the development pipeline in similar fields such as Architecture and Construction, most of the generated knowledge is implicit in nature since many aspects of it is based on experience. Due to this nature of knowledge generation, it becomes very important to preserve such knowledge to ensure that future members of the game development organization can have adequate resources to continue to function at the highest efficiency. Another issue with game development organizations is many a times there may occur scenarios where teams may need to switch their focus and work on a single project, which is different from their specialty. [5]. There are also cases where many experts within the organization leave, leading to a vacuum in the organizational knowledge. [21] This vacuum can prove to be detrimental because new members may not be able to adequately get assistance settling into the inner workings of the organization. At the same time, it takes a lot of time, money and effort to train these individuals back up to the caliber that the previous experts possessed, thus needing more investment of resources. For such an unpredictable field such as game development, where trends make and break overnight [6, 7] such investments may take away the focus from development processes, thus becoming a roadblock on the path to success for the organization.

This is the purpose of this project. To investigate a way to manage implicit knowledge generated in such an unpredictable and rapidly developing ecosystem that exists within a game development organization, in efforts to establish guidelines to better structure and organize such knowledge.

2. Review of Literature

Just as Nonaka and Takeuchi (1995) [3] provided a basis for the creation of knowledge types for organizations, previous research details the viewpoint of tacit (or implicit) knowledge in general. Polanyi [8] in his research attributed implicit knowledge to have a highly personal and context-based knowledge. That research also specifies the highly informal nature of implicit knowledge, thus only through very specific methods these may be communicated to someone else.

This project focuses on the creation of a framework and establish some guidelines, through which implicit knowledge that is generated within a game development organization can be recorded, preserved, and accessed by the organization, effectively turning it into organizational knowledge. As mentioned in Section 1 and Section 2, knowledge is a vital part of an organization which is built upon and extended through its members' experiences, schemas, and understanding. Thus, since implicit knowledge is such an important, mostly untapped resource, the purpose of this (work-in-progress) project is to explore a way to integrate this implicit knowledge into existing knowledge base of the organization to have a more dynamic and detailed knowledge base [4].

This framework proposes four major components that will work in tandem to form a system that takes it closer to the objective. The components are listed as follows:

1. Knowledge Base (and Knowledge Graph)
2. Expert System
3. Community of Practice and Socialization Component

Each component has a specific function, and exists as a feedback loop process, that works towards the goal of enhancing organizational knowledge.

2.1 Knowledge Base (and Knowledge Graph)

An important factor in organizing the knowledge generated within an organization is the development of a knowledge base. According to Nguyen (2020) [9] knowledge bases (KBs) are representative of the relationships between different entities, which exist in the form of triples (i.e. a three word triplet). The first word and the third word are connected contextually by the second word (which describes the relationship between the first and third word). The author emphasizes for the need to have embedding vectors, to be able to represent the information and relationships with a more manageable number of dimensions. One of his suggested directions knowledge bases could take was to understand the behavior of knowledge bases with readily available data in the form of triples. Classification of tacit knowledge in such triple categories may show an indication towards it being recorded and analyzed effectively.

Woo et. al. (2004) [4] in their research also talk about the problems that arise during conversion of implicit knowledge to explicit. According to them, numerous difficulties arise in extending and enhancing knowledge bases, due to the existing structure already being implemented and not allowing a lot of flexibility with the explicit content within the system. Another issue is that good knowledge bases require a lot of input from experts, whose knowledge primarily comprises of implicit knowledge. [10, 11] In game development, knowledge is generated in every stage, thus contributing the need for it to be recorded and preserved into a knowledge system. These issues are also applicable to the game development industry, and thus this paper focusses on an attempt to give some structure and guidelines towards being able to utilize this untapped resource.

Further confirmations that there exists a relation between the existence of a proper knowledge base and the growth of a firm has been recorded by Grillitsch et. al. (2018). [12] In their research, the conclusion was that having a combinatory knowledge base system (that utilizes implicit, explicit, and organizational knowledge) can lead to the growth of an organization as well as lead to further innovation. This effect is seen in to have a significant presence when the organization has a unique product to offer, as well as needing to have a competitive advantage – two conditions that are prevalent in the game development sector. [13, 14] One point they note is that their results were inconclusive for large firms (due to the high volume of different knowledge types in the system; 250 employees or more). This might be something to note even in the game development sector. Zhang and Baden Fuller [15] in their research, investigated the importance of having a technological knowledge base in an organization, and how it could affect the way the organization is structured and could collaborate with existing technologies. They stated that having a larger knowledge base provides organizations (primarily from the pharmaceutical, chemical, and agro-food) with the confidence and ability to share knowledge and prefer alliances with similar companies. [14] This promotes inter-organization knowledge sharing and expansion of existing knowledge base.

Knowledge bases can be visualized through a technique known as knowledge graphs or knowledge maps. These techniques result in ways that can depict a representation of the entities and relationships that exist in the knowledge base. The primary use for knowledge maps is to provide a convergent thinking schema to locate the entities that can be associated with the concepts that people are seeking. These are mostly static and lay out a roadmap of sorts, to navigate one's way from a start point to an end point. This can prove to be a bit of a drawback when it comes to knowledge maps [4]. Research also shows the importance and implications of having a dynamic knowledge map. [4, 16, 17] A key advantage is that having a dynamic knowledge map can contribute towards faster knowledge transfer within organizations. The key component of a dynamic knowledge map is its dependence on the world wide web, which can function as a knowledge base. This is further supported by the work of Talmor and Berant (2018) [18]. Their research investigated towards considering the implications of using the world wide web as a resource or tool to answer complex questions. The dataset they developed is targeted towards applications in semantics, and compositionality, as well as be able to get solutions from a combinatory knowledge source, to improve complex question-answer performance. The takeaway from this could prove helpful to parse implicit knowledge texts, which are usually penned down as interview responses. [19]

2.2 Expert System

Expert systems can be defined as applications that mimic the reasoning, thinking, and decision-making processes of a human expert [22,23]. In their initial research, Singh et. al. (1996) described the importance of expert systems to facilitate in improved performance and information access and management accuracy, when applied on well-developed information systems. Their major goal is to tackle complex tasks, and decision-making processes, to provide a helping hand to the user as and when required. According to Asemi et. al. (2020) [23] expert systems consist of two main elements: knowledge base and inference engine. Knowledge base has been extensively discussed in Section 2.1 and thus, its importance is utmost. Information can be extracted from the knowledge base through two phases: “if” (conditions) and “then” (results) phases. The inference engine is responsible for the reasoning aspect of these intelligent systems. This component parallels the human brain when it comes to decision making, based on various constraints and rules.

The main aspect of expert systems was discussed further by Devadson and Vespry (1996), [24] when the system they introduced (which focused mainly on library specialization systems) was used to determine different applications to improve recommendations for library systems. Applications such as “Knowledge-

based indexing” [23], “Natural Language Processing and Abstracting” [25], Cataloging [23], and Online information retrieval [26] are some of the important applications that was provided as an output from the system. These are the important topics that this paper’s proposal would like to consider into its expert system. They concluded that having a user’s information behavior is an appropriate way to design intelligent systems and their development can be further enhanced using intelligent agents and having user profiles (which will be further touched upon in Section 2.3).

Asemi et. al. (2020) [23] in their research have consolidated a review of various authors and their contributions towards this field of application of expert systems, which proved to be valuable for understanding the framework this paper is trying to propose. Hwang et. al. (2011) [27] proposed a mind tool that showed grid-based knowledge mapping. Their objective was to help organize and share knowledge among students for various field observations i.e., the implicit knowledge that was recorded. Another research by Ismail and Kareem (2013) [28] focused on novices searched, located, and chose resources to seek specific information about a topic. Similarly, McDonnell and Shiri (2011) [29] investigated information retrieval systems and the interaction with user interfaces when it comes to the world wide web and consumer behavior.

Gu et. al., 2019 [30] developed an expert system using machine learning and big data techniques, where they tried to predict whether a National Hockey League (NHL) team will win a game. They list the primary components of their system to have a knowledge base comprised of various web sources, “principal component analysis, nonparametric statistical analysis, a support vector machine (SVM) and an ensemble machine learning algorithm”. [30] The proposal in this paper also tries to draw inspiration from such research to further the direction in using expert systems in the field of game development.

All these examples can illustrate the importance that expert systems can have when dealing with knowledge management and knowledge transfer. Expert systems have been used over various technologies to assist in information management and help users decide future decisions and involving it to streamline and manage knowledge management and transfer within game development organizations.

2.3 Community of Practice and Socialization Component

According to Mazorodze & Buckley (2020) [42] Communities of Practice (CoP) are the most efficient and effective knowledge transfer tool in an organization with intensive knowledge generation regimes. This component has three components – the domain, the community, and the practice itself – all of which work in tandem to contribute towards organizational knowledge. This concept of using Communities of Practice as a knowledge transfer tool was further outlined by Wenger et. al., 2002 [43] where they stated the seven principles for effective development of CoP – designing for evolution (due to the organic nature of these communities), allowing internal and external dialogue perspectives, inviting participation (from all levels of expertise), developing private and public discussion and knowledge sharing spaces, focusing on the value provided by its members, combining everything into a familiar idea that incites excitement, and creating a rhythm for the community to explore, learn, and grow. O’Toole [44] in his research found that intra-organization knowledge sharing and team members leveraging each other’s knowledge resources could lead to less chances of abandoning projects. Another conclusion he made is that through such experiences of members of the organization, it could potentially lead to understanding where the organization lacks and performance failures.

Cohendet and Simon (2007) [45] in their research also state the concept of communities of specialists in the field of game development (video game development firms). These communities include game developers, software programmers, 3D artists, script writers etc. who contribute to the ecosystem of game

development pipelines within a studio. According to their research this rich and diverse technical background of team members as well as their cultural backgrounds can prove to be a major balancing task. To combat these, they suggest some steps organizations specific to game development could take to enhance the internal communities' creativity. One of the steps is to develop a hybrid form with hierarchies and roles defined based on each specialist's skillset. Rather than have a top-down approach to all projects (where one lead takes decisions and the subordinates follow), it is advised that the communities work with each other, keeping the problem at the center. Shared working space also can contribute to a more dynamic vision for each project. One of the key takeaways is that the friction, competition, and improvisation between these creative internal communities can lead to organizational innovation.

Similarly, Burger-Helmchen and Cohendet's (2011) [46] research takes a deep dive into the different types of communities related to the video game industry. Internal communities are not the only source for developing implicit knowledge (and in turn organizational knowledge) within a game development ecosystem. Outside communities i.e., users too enable the internal communities to come up with novel solutions to problems that contribute to the organizational knowledge base. They categorize the userbase of video games into three categories: player communities (primarily the consumers), tester communities (usually to evaluate the products) and developer communities (primarily concerned with developing and producing elements of the game). Their results concluded that having these communities introduced much-needed feedback into the development process of the organization (corroborated through interviews with specialists within the organization), thus noting the importance of the influence of developer communities.

Along these lines, Mushtaq et. al. (2018) [47] explored the mining of implicit (and explicit) knowledge in crowdsourced communities in the technology sector. They compared major resource pools (such as StackOverflow) which are primarily unstructured but heavily used and tried to classify and structure the implicit and explicit knowledge that exists. In this way, the idea is to leverage the implicit knowledge of experts from these communities. Similar logic can be applied to internal members of an organization, which would enable to strengthen the organizational structure and knowledge. There certainly exists evidence that communities of practice are untapped resources for implicit knowledge. MacFarlane et. al. (2020) [48] discusses the potential of using machine learning to retrieve this information and knowledge. They state the importance of text search through search engines and how knowledge organization can be optimized through these techniques. However, they do state the existence of semantics gap, which can be overcome through techniques of contextual tagging and supervised learning techniques (for multimedia information).

3. Case Study: Visualizing a Game Postmortem

To understand how the expert system visualization component for managing implicit knowledge within a company may work, this project tried to shed some light towards how a realistic scenario of handling such knowledge might look like. The focus of this paper is thus mostly on the visualization component (both from a knowledge graph and interaction system point-of-view). From previous literature, stated in Section 2, the project adopts many features seen in knowledge bases, expert systems, and a social network driven dynamic information management systems.

Some assumptions were made regarding how to proceed, and since this project has been exploratory in nature, the aim was to gather insight from existing data. Since there were not many formal models for game development taxonomy and ontology, a very basic taxonomy and ontology was developed, for a couple of steps in the development process. The idea here is that, if there does exist a way to visualize game development process information right now via such assumptions, then there exists enough evidence to further investigate and iterate along these lines to reach the end goal.

3.1 Dataset

To start off on a common point, it was identified that one of the most important tools in game development are postmortems. [31, 32, 33] Postmortems essentially are summaries of the entire development cycle for a specific video game, done by the developers themselves. The main objective of a postmortem is to highlight the issues and notable events that occurred throughout the development cycle, and how these issues were addressed and tackled. It generally is a very detailed description that can give readers an idea of the challenges that can come up during development of video games and the multitude of factors that influence the course of development.

The dataset used was the “Dataset of Game Development Problems” by Politowski et. al., 2020 [33]. This dataset investigated a total of 200 postmortems for video games, and manually coded the problems and issues encountered. The timeline of games covered was 1998 to 2018 (twenty years). Towards the end, they were able to conclude 1035 problems from all the postmortems. They categorized all these problems into 20 different types and classified them as a part of either of the three groups: Production Problem Group, Management Problem Group, and Business Problem Group. Figure 2 shows a consolidated view of the total number of problems per type, that was found. It was found that a total of 45.22% of the problems were Management based, 45.02% problems were Production based, and 9.76% were business based.

3.2 Important Dimensions and Tools Used

For the scope of this study, the number of dimensions being considered are of three types: technical problems, team problems, and documentation problems. This is to reduce the number of variables for initial analysis, and to streamline the input data. Here, technical problems represent explicit knowledge, team problems represent implicit knowledge, and documentation problems represents organizational knowledge. In this way, the Community of Practice component gets addressed, because of the assumption that the system is being developed from the point of view of the programming team. Another advantage to this is that now the taxonomy will be built around the programming department, with certain keywords and information that is important to that specific domain.

The tools used for the analysis of this dataset was Tableau (to visualize and analyze data) and Python (for development).

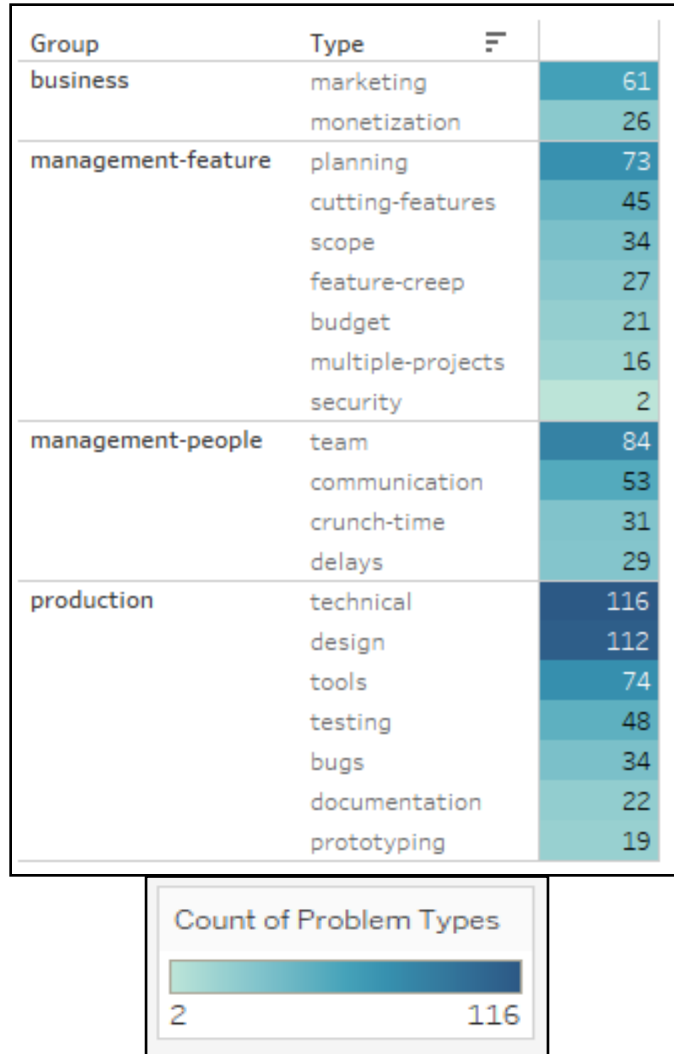


Figure 1: Summary of dataset. Darker color means a greater number of problems (extracted from postmortems)

3.3 Methodology

One of the most important ways to represent knowledge in the form of a knowledge graph was laid down by Nickel et. al., 2015 [34]. They introduced the concept of converting natural language-based sentences (that humans use in regular speech), to triples. They stated that every sentence can be broken down into a “subject-predicate-object” triple (SPO triples). Subject and object are the entities in this context, and the predicate is the relationship that exists in-between the entities. In this way, an ontology is created for the current taxonomy, and that enables the creation of a visual network graph to convey information. Using this logic, a preliminary test case was developed.

Nickel et. al., (2015) [34] used the example sentence “*Leonard Nimoy was an actor who played the character Spock in the science-fiction movie Star Trek*” to construct the triples (shown in Table 1). Based on that, they created a knowledge graph to establish a visual flow diagram between entities and relationships. Using Python (Spacy and networkX modules) [35, 36, 37] the first step was an attempt to

recreate the flow diagram through code. To make it simple, directionality was not considered in the final output (Figure 2).

Table 1: Subject-Predicate-Object Triples for the example sentence

Subject	Predicate	Object
LeonardNimoy	Profession	Actor
LeonardNimoy	starredIn	StarTrek
LeonardNimoy	Played	Spock
Spock	characterIn	StarTrek
StarTrek	genre	ScienceFiction

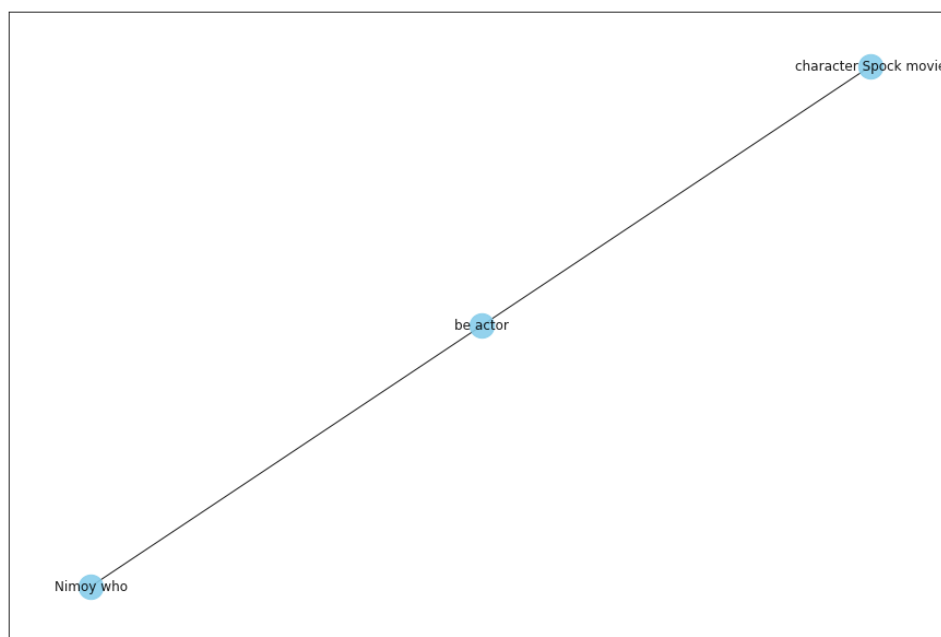


Figure 2: Custom code to visualize the sentence (two triples)

This showed somewhat promising results that SPO triples could be represented as digital networks that would show the relationships between entities. However, there are a lot of inaccuracies in the graph, and certainly it can be improved over time. One such improvement would be to try and have semantically correct nodes and show the direction of flow of information. Next, Politowski's [33] research was considered, and one random game postmortem was selected to be analyzed. A lot of these were done through interviews and recorded in the manner resembling human speech, thus providing for a good input for the natural language processing algorithms to work on. Figure 3 illustrates an example postmortem which was selected for analysis, and manually color coded (based off on a game developer's experience of what information could be extracted from it).

“Many of the systems we inherited were only partially implemented and virtually undocumented. To make things worse, much of the code was written to implement advanced animation and mathematical methods used all over the game engine, and we didn't have a technical design document that we could refer to in order to determine the intended solution for these systems.”

Figure 3: Example post-mortem. Color coded (4 colors) to represent four major problem areas (derived from implicit knowledge)

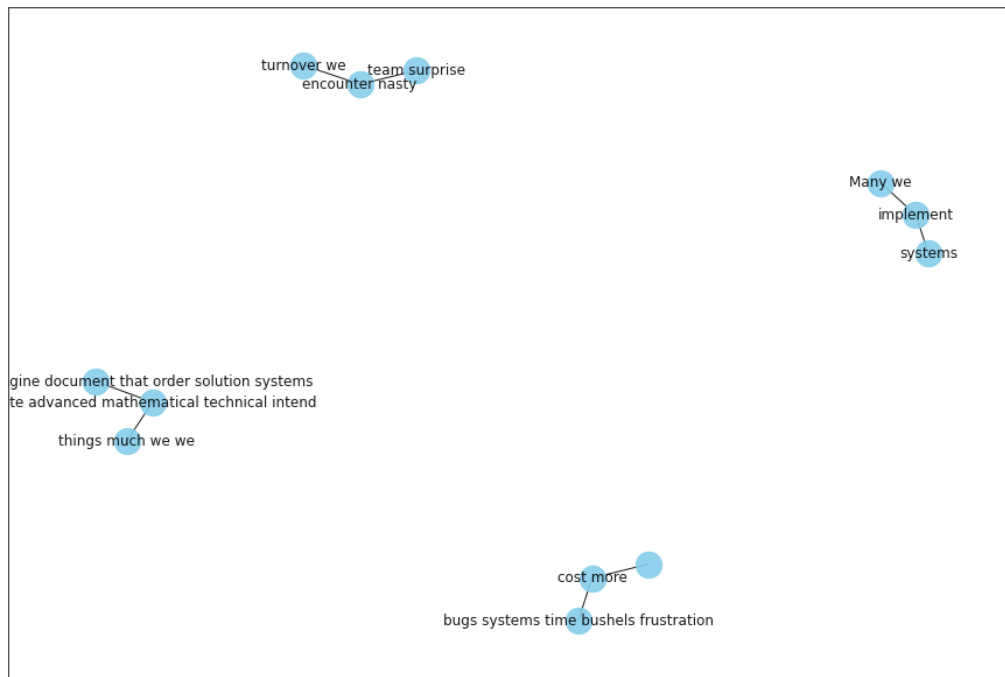


Figure 4: Network for example postmortem (no processing done)

Using the same technique applied to the previous example sentence, this short text was analyzed, and the output graph can be seen in Figure 4. This is not a good graph, as it has several pitfalls. Visually unattractive, no insight can be drawn from it, extra artifacts being added in, all contribute towards it not being a helpful visualization. The next step would be to extract meaningful information out of this body of text, since converting it into triples simply does not establish enough context for the algorithm to understand how to represent this information visually. For information extraction, a method called Tokenizing was used, which essentially can be used to find unique elements from a text block [35, 38, 39]. Using Python (Gensim module) [40, 41], the block of text was analyzed, and converted into a dictionary of unique words (or entities).

These unique words did make the task of the text analysis easier, however the meaning of the context was lost, as well as the rearrangement of the individual words might have resulted in the text being analyzed differently (especially since there was no robust comparator dataset focused on game development

taxonomies and ontologies to make sense out of it). A similarity score analysis as well as two-dimensional graph plot of the words, displays similar words clustered together, and that might hint towards the idea that there was some meaning lost in translation due to tokenization. As seen in Figure 5, the plot places the words all over the similarity plot, thus implying the need for us to sort out specific terms related to the knowledge that needs to be preserved.

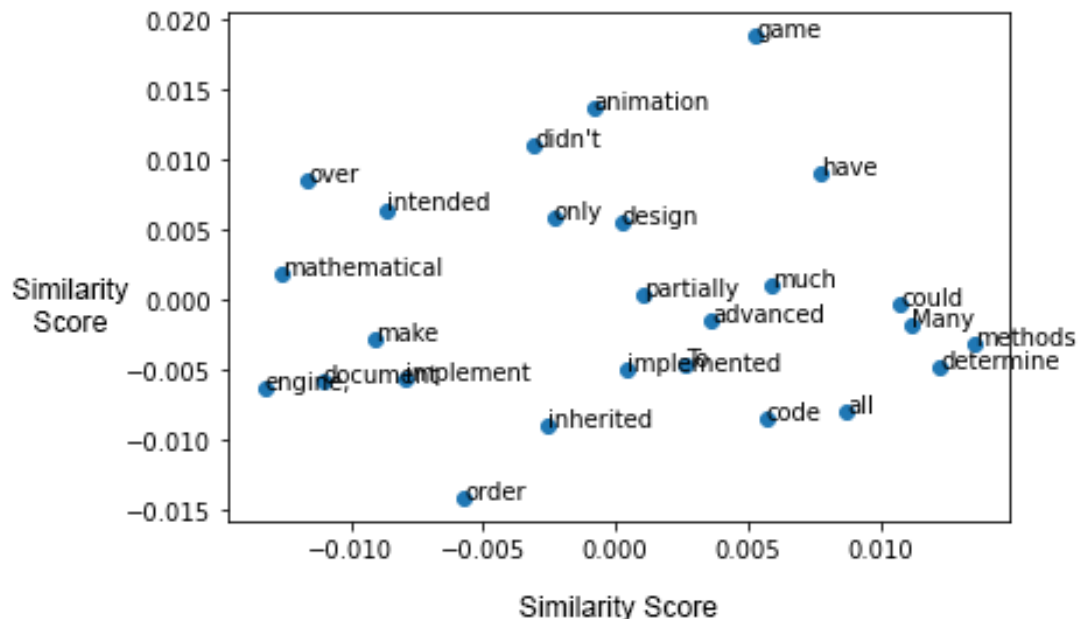


Figure 5: Two-dimensional similarity score plot (after tokenization)

This is where this project attempts to provide a direction through some basic trial and error stages. Due to the lack of formal taxonomy, there is not a robust approach to this. Thus, an approach that was settled on where two filter lists were developed. The first list comprised of words that human speech has for context clues and does not contribute to the meaning for the text individually. This list removes the unwanted words, assuming in the process the context is driven more towards game development. The second filter list was a custom assumed taxonomy of words related to the task of programming in game development i.e., a list of common words that may be used when programmers are working on a video game. This list was developed primarily from implicit knowledge of the authors as well as from the various other postmortems and was used to search for terms specific to a domain. These two lists were added into the text analysis algorithm, and another similarity score plot was created, illustrated in Figure 6.

This approach seemed to work to an extent, since now only those words remained that indicated towards the major problem areas that the postmortem referred to. The plot is also much clearer and we can see that each of the four topics lie on four different quadrants (divided if lines are drawn through the origin). This could indicate towards the start of a layout, that if provided with a more complex postmortem, and a more robust list of video game development taxonomies and ontologies. Using these target words, the next step was to be able to visualize the postmortem in a way that conveys the same information. The same algorithm used to visualize the sentence used in Nickel et. al. (2015) was used, however initial results indicated that just having the target words would not produce an optimal outcome. Thus, there existed the need to have proper input to the visual network algorithm. Through more trial-and-error processes, a block of text was generated that could convey the same information in the original postmortem (Figure 7).

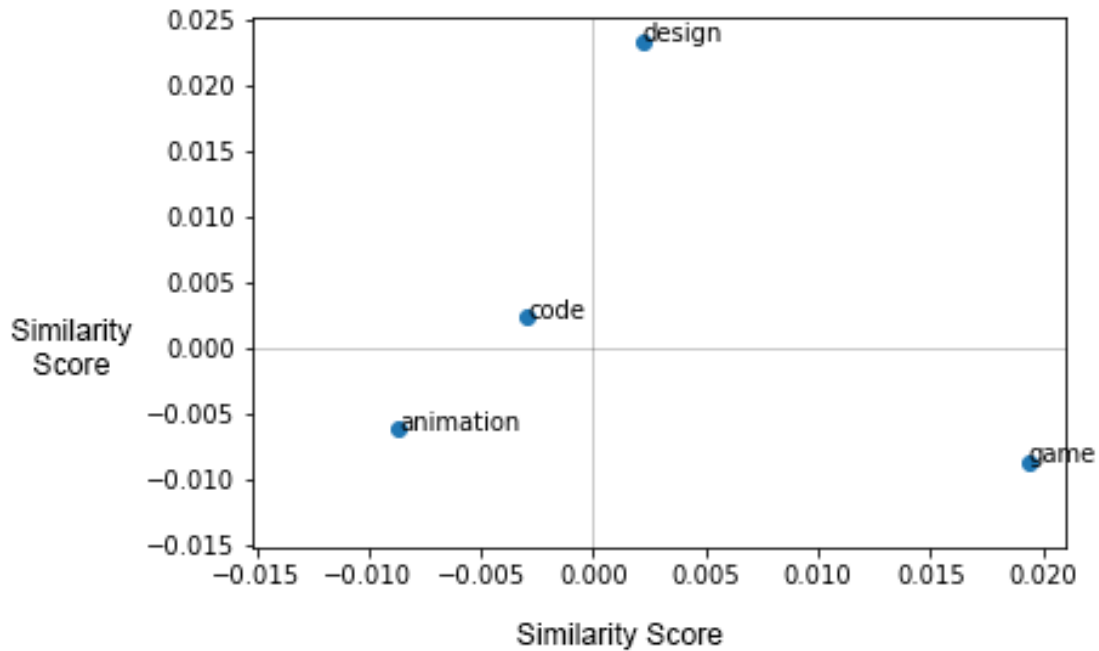


Figure 6: Two-dimensional similarity score plot (after tokenization and filtering)

*“Code undocumented. Game engine complex.
Animation system advanced. Design document
missing.”*

Figure 7: Refined postmortem to be used as input for network graph. Color coded (4 colors) to represent four major problem areas (derived from implicit knowledge)

A lot of these trial-and-error stages involve working with implicit knowledge that the authors had acquired over the years of working as game developers and visualization engineers, thus to some extent, illustrating how implicit knowledge in the minds of experts comes into play when there may be a lack of information. The final step is to take this revised postmortem and analyze it using the initial network algorithm. Figure 8 is the result.

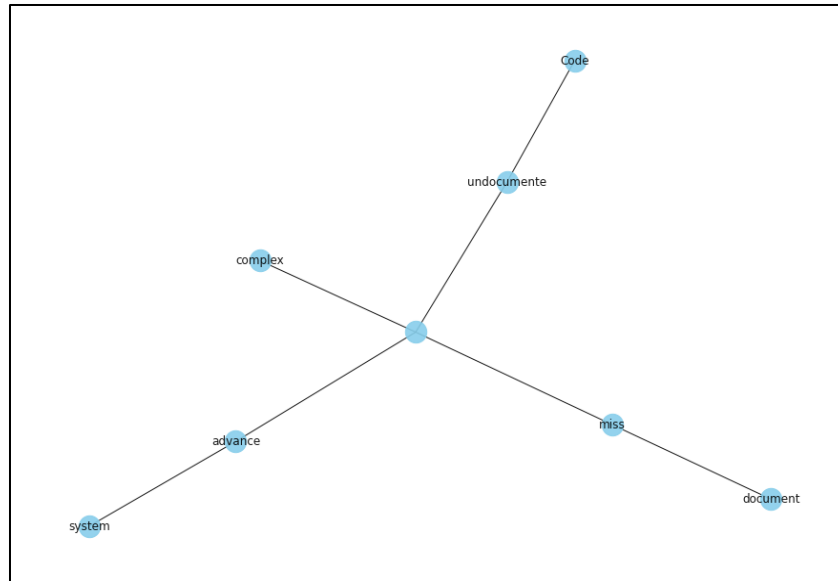


Figure 8: Network graph (visualized knowledge graph) for refined postmortem

From this network graph, there exists a much clearer and more readable visual graphic, that conveys the information contained in the original postmortem. As a note, throughout the entire process a lot of is dictated by trial and error, which was driven by implicit knowledge and experience. This could play out to the advantage of the final framework, since there is no real structure or syntax that can clearly distinguish implicit knowledge from explicit knowledge, and this project is attempting to provide some structure to how this knowledge can be recorded, stored, and accessed. Figure 9 represents with added dimensions (coordinate in 2-D space as well as weighted line width to represent the strength of the relationship, what the final interactable graph could look like).

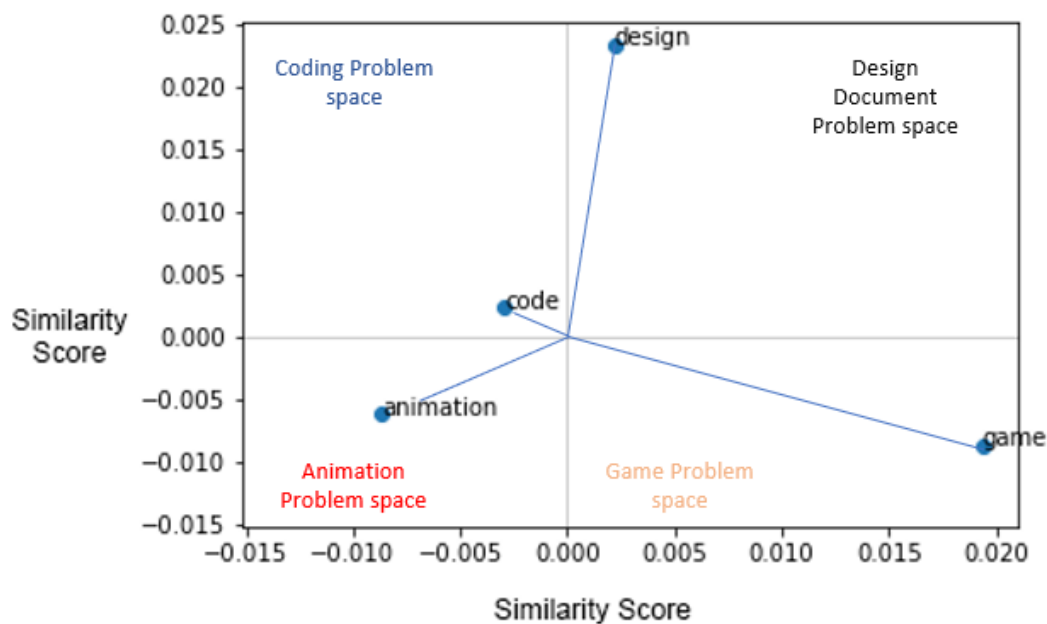


Figure 9: Network graph with possible extra dimensions (4 color codes from postmortem)

4. Conclusions

Thus, from this small exploratory visualization, there does exist a potential in trying to parse informal texts (such as game postmortems) into effective visualizations, as well as have parameters attached to them, that could assist in the construction of an interactive and dynamic knowledge base. With proper input of data, refined networks can be made to establish a visual hierarchy as well as show the relationships between major components that exist within the knowledge base. In this way, there will exist a way for organizations to view their accumulated knowledge as well as the members that play an essential role in maintaining and interacting with each of their specific domains, in a formal and structured way.

There are also major benefits to developing a structured knowledge base as it would allow the organizations to expand it through machine teaching techniques so that there is continual updating of the generated organizational knowledge. That would also support the point in the literature review of companies having enough confidence to share their knowledge with other organizations as well as be more open towards inclusion of new members and technologies in their workflow.

In terms of game development, these techniques could prove to be able to provide a structure to the knowledge generated by the individuals, as well as by the collective members of the studio. Since the game development industry needs to deal with a lot of technical changes as well as members shifting from one team to another to meet deadlines, it would be beneficial to have a system that works as a passive record of all these changes and experiences, sorts them out to proper domains, and records and preserves all this information in an accessible way, to promote the growth of the organization and help create libraries of information that can help in transfer of knowledge – be it between organizations (organizational knowledge), organizations and members (explicit knowledge), or in between members (implicit knowledge).

5. Limitations

Since this paper was exploratory in nature, and the project is still a work-in-progress, many assumptions had to be made to try and attempt to formulate a solution that would be viable. Thus, this project does have some limitations, that as the iterations progress, will be addressed.

Since there is no formal taxonomy and ontology that exists for the domain of game development, a lot of information regarding the development cycles must be obtained from postmortems, reviews, and implicit knowledge (i.e., draw upon the experience of people as game developers). Thus, this approach may not always yield the best results nor be applied to all cases, until a solid foundation comprising of a generalizable taxonomy and a competent ontology have been developed. This is also a reason why the output graphs are not usually semantically accurate and clean.

In this paper, for testing purposes, the comparator databases used for natural language processing (NLP) and word embedding or word vectorization may have played a role in not accurately being able to properly contextualize the key terms with the field of game development, without some procession being done on the original input. Thus, there may be cases where there will be removal of vital information from the source material due to the process. Thus, this process needs to be more robust and properly evaluated so that it leads to a more generalizable solution, as well as be extended to other domains. The natural language processing dataset used for comparison too was originally developed for Biomedical NLP. [36]

A lot of the stages in this experimental and exploratory research involves trial and error to establish an objective. Thus, the idea is over time to refine the methodology and end up with a clearer output.

6. Future Work

Since this initial experiment was to try and establish whether there is a possibility for developing ways to convert implicit knowledge to explicit, in a way that benefits organizations and its members. There still is a lot of improvements and additions that need to be done, for this to achieve a state that can be implemented in a working environment.

First, the most important thing is that we formalize the way that communication occurs within game development ecosystems, to be able to identify important concepts and establish a structured information flow. For this, development of a common taxonomy of words and concepts that encompass most of the stages of development and its components. Once a reasonable taxonomy has been established, to develop a good knowledge base, there needs to exist relationships between these concepts, thus creating an organizational ontology. This would constitute a knowledge base, that would be ready to have functions (such as machine teaching) applied to them for further expansion in applications.

The next important step is to integrate or introduce an expert system in the workflow, that takes advantage of this completed knowledge base. Having an expert system would enable the users to be able to get recommendations based on their need, by working along with the knowledge base system to deliver specific components that the user would like to work with. This component would contribute towards a dynamic documentation system as well as a dynamic knowledge base, that would allow more flexibility to an organization. This would also ensure a visualization component is present and gives more visual based feedback for the network of structured organizational knowledge.

Finally, through techniques of Community of Practice, it would enable to promote a shared platform for knowledge transfer, as well as allow experts to unload their implicit knowledge in a more refined form, to be accessed by anyone who desires.

All of this has been mapped out in a roadmap, in the hopes that this framework will allow game development organizations to retain information in a more structured way, long term.

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